

# Dimitris Kolonelos

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**Contact:** dimitriskolonelos@gmail.com, dimitris.kolonelos@berkeley.edu

**Website:** <https://dimkolonelos.github.io/>

**INTERESTS** Cryptography, Succinct Cryptographic Primitives, Zero-Knowledge Proofs, Authenticated Data Structures, Advanced Encryption Schemes, Blockchain Applications.

**EXPERIENCE** **Postdoc** *September 2024 - now*  
UC Berkeley  
Host: Sanjam Garg

**Visiting Scholar** *June 2023 - July 2023*  
UC Berkeley  
Host: Sanjam Garg  
Work on: *SNARKs, Threshold Encryption.*

**Research intern** *April 2021 - August 2021*  
Ethereum Foundation  
Host: Mary Maller  
Work on: *Zero-Knowledge Proofs over highly untrusted settings (subverted RSA groups).*

**Research intern** *September 2018 - February 2019*  
IMDEA Software Institute  
Host: Dario Fiore  
Work on: *Efficient Zero-Knowledge Proofs for privacy-preserving applications.*

**Undergraduate Research Assistant** *September 2017 - July 2018*  
NTUA Computation and Reasoning laboratory (Corelab)  
Advisor: Aris Pagourtzis  
Work on: *Anonymous Survey Systems through cryptographic techniques. Improving privacy of 'Anonize', an existing anonymous survey system.*

**EDUCATION** **PhD in Computer Science** *February 2019 - February 2024*  
IMDEA Software Institute & Universidad Politécnica de Madrid, Spain  
Advisor: Dario Fiore  
Thesis: *Succinct Cryptographic Commitments with Fine-Grained Openings for Decentralized Environments.*

**MEng Electrical and Computer Engineering (5-year)** *Sept 2011 - Jul 2018*  
National Technical University of Athens (NTUA), Greece

**SHORT VISITS** IMDEA Software Institute (September 2025)  
Host: Dario Fiore

Bocconi University (August 2025)  
Host: Giulio Malavolta

The Chinese University of Hong Kong, Hong Kong (December 2023)  
Host: Sherman S. M. Chow

IRIF, Paris (November 2023)  
Host: Geoffroy Couteau

Microsoft Research, Redmond (November 2022)  
Host: Melissa Chase & Esha Ghosh

Max Planck Institute for Security and Privacy (MPI-SP), Bochum (February 2022)  
Host: Giulio Malavolta

## AWARDS

**Berkeley Center for Responsible, Decentralized Intelligence (RDI) Fellowship.** (February 2025 - April 2025)

**Protocol Labs research gift:** award of one-year PhD funding (September 2019 - August 2020)

## PUBLICATIONS

*Jigsaw: Doubly Private Smart Contracts*

Sanjam Garg, Aarushi Goel, Dimitris Kolonelos, Rohit Sinha  
IEEE S&P 2026

*A Framework for Witness Encryption from Linearly Verifiable SNARKs and Applications*

Sanjam Garg, Mohammad Hajiabadi, Dimitris Kolonelos, Abhiram Kothapalli, Guru-Vamsi Policharla  
CRYPTO 2025

*Split Prover Zero-Knowledge SNARKs*

Sanjam Garg, Aarushi Goel, Dimitris Kolonelos, Sina Shiehian, Rohit Sinha  
PKC 2025

*Threshold Encryption with Silent Setup*

Sanjam Garg, Dimitris Kolonelos, Guru-Vamsi Policharla, Mingyuan Wang  
CRYPTO 2024

*Zero-Knowledge Proofs for Set Membership: Efficient, Succinct, Modular*

Daniel Benarroch, Matteo Campanelli, Dario Fiore, Kobi Gurkan, Dimitris Kolonelos  
Designs, Codes and Cryptography, Volume 91

*Cuckoo Commitments: Registration-Based Encryption and Key-Value Map Commitments for Large Spaces*

Dario Fiore, Dimitris Kolonelos, Paola de Perthuis  
ASIACRYPT 2023

*Distributed Broadcast Encryption from Bilinear Groups*

Dimitris Kolonelos, Giulio Malavolta, Hoeteck Wee  
ASIACRYPT 2023

*Efficient Registration-Based Encryption*

Noemi Glaeser, Dimitris Kolonelos, Giulio Malavolta, Ahmadreza Rahimi  
ACM CCS 2023

*Efficient Laconic Cryptography from Learning With Errors*

Nico Döttling, Dimitris Kolonelos, Russell W. F. Lai, Chuanwei Lin, Giulio Malavolta, Ahmadreza Rahimi

EUROCRYPT 2023

*Zero-Knowledge Arguments for Subverted RSA Groups*

Dimitris Kolonelos, Mary Maller, Mikhail Volkhov

PKC 2023

*Succinct Zero-Knowledge Batch Proofs for RSA Accumulators*

Matteo Campanelli, Dario Fiore, Semin Han, Jihye Kim, Dimitris Kolonelos, Hyunok

Oh

ACM CCS 2022

*Ring Signatures with User-Controlled Linkability*

Dario Fiore, Lydia Garms, Dimitris Kolonelos, Claudio Soriente, Ida Tucker

ESORICS 2022

*Inner Product Functional Commitments with Constant-Size Public Parameters and Openings*

Hien Chu, Dario Fiore, Dimitris Kolonelos, Dominique Schröder

SCN 2022

*Zero-Knowledge Proofs for Set Membership: Efficient, Succinct, Modular*

Daniel Benarroch, Matteo Campanelli, Dario Fiore, Kobi Gurkan, Dimitris Kolonelos

Financial Cryptography and Data Security 2021

*Incrementally Aggregatable Vector Commitments and Applications to Verifiable Decentralized Storage*

Matteo Campanelli, Dario Fiore, Nicola Greco, Dimitris Kolonelos, Luca Nizzardo

ASIACRYPT 2020

## TALKS

*Jigsaw: Doubly Private Smart Contracts*  
IOG Seminar, Online, October 2025

*Jigsaw: Doubly Private Smart Contracts*  
IMDEA Software Institute, Madrid, September 2025

*Threshold Encryption with Silent Setu*  
Bocconi University, Milan, August 2025

*Jigsaw: Doubly Private Smart Contracts*  
The Science of Blockchain Conference 2025 (SBC 2025), Berkeley, August 2025

*Jigsaw: Doubly Private Smart Contracts*  
Lagrange Science Fair, Online, August 2025

*Jigsaw: Doubly Private Smart Contracts*  
o1Labs Reading Group, Online, July 2025

*Split Prover Zero-Knowledge SNARKs*  
Public Key Cryptography 2025, Roros, May 2025

*Jigsaw: Doubly Private Smart Contracts*  
Workshop on Cryptographic Tools for Blockchains, Madrid, May 2025

*Jigsaw: Doubly Private Smart Contracts*  
Bay Area Crypto Day, Sunnyvale, April 2025

*Beyond Garbling: Efficient Advanced Encryption Schemes without Trusted Authority*  
Eurocrypt 2024 Workshops: Theory and Practice of Laconic Cryptography, Zurich, May 2024

*Registration-Based Encryption: How to build it without garbling*  
The Chinese University of Hong Kong, Hong Kong, December 2023

*Distributed Broadcast Encryption from Bilinear Groups*  
Asiacrypt 2023, Guangzhou, December 2023

*Cuckoo Commitments: Registration-Based Encryption and Key-Value Map Commitments for Large Spaces*  
Asiacrypt 2023, Guangzhou, December 2023

*Efficient Registration-Based Encryption*  
ACM CCS 2023, Copenhagen, November 2023

*Registration-Based Encryption: How to build it without garbling*  
IRIF Crypto Reading Group, Paris, November 2023

*Distributed Broadcast Encryption from Bilinear Groups*  
Stanford Security Seminar, Palo Alto, August 2023

*Distributed Broadcast Encryption from Bilinear Groups*  
UC Berkeley Cryptography Seminars, Berkeley, June 2023

*Zero-Knowledge Arguments for Subverted RSA Groups*

Public Key Cryptography 2023, Atlanta, May 2023

*Succinct Zero-Knowledge Batch Proofs for RSA Accumulators*  
Microsoft Research, Redmond, November 2022

*Succinct Zero-Knowledge Batch Proofs for RSA Accumulators*  
Crypto Economics Security Conference (CESC) 2022, Berkeley, October 2022

*Succinct Cryptographic primitives with applications to the Blockchain*  
Cybersecurity Research Network meeting, Lleida, March 2022

*SoK - Vector Commitments*  
Ethereum Foundation, Online, June 2021

*Zero-Knowledge Proofs for Set Membership: Efficient, Succinct, Modular*  
Financial Cryptography and Data Security 2021, Online, March 2021

*Zero-Knowledge Proofs for Set Membership: Efficient, Succinct, Modular*  
Monash Cybersecurity Seminars, Online, February 2021

*Incrementally Aggregatable Vector Commitments and Applications to Verifiable Decentralized Storage*  
Asiacrypt 2020, Online, December 2020

*Incrementally Aggregatable Vector Commitments and Applications to Verifiable Decentralized Storage*  
Protocol Labs Research Seminar Series, Online, November 2020

*Vector Commitment Techniques and Applications to Verifiable Decentralized Storage*  
Theory and Practice of Blockchains (TPBC) 2020, Online, July 2020

*Zero-Knowledge Proofs for Set Membership: Efficient, Succinct, Modular*  
Theory and Practice of Blockchains (TPBC) 2020, Online, June 2020

*Zero-Knowledge Proofs for Set Membership: Efficient, Succinct, Modular*  
Crypto Economics Security Conference (CESC) 2019, Berkeley, October 2019

**SERVICE**                      **Program Committees:** ACM CCS 2026, Usenix Security 2026, IEEE S&P 2026, ACM CCS 2025

**External Reviews:** EUROCRYPT 2026, TCC 2025, CRYPTO 2025, EUROCRYPT 2025, TCC 2024, ASIACRYPT 2024, CRYPTO 2024, EUROCRYPT 2024, TCC 2023, ASIACRYPT 2023, EUROCRYPT 2023, ACNS 2023, CRYPTO 2022, PKC 2021, ASIACRYPT 2021, EUROCRYPT 2021, FC 2021, ACM CCS 2020, PKC 2020

**LANGUAGES**                Greek (native), English (Proficiency), Spanish